

CMI and Cytokine Profiles in Leprosy

These clinical presentations correlate with the level of cell-mediated immunity (CMI) against *M. leprae*. The standard measure of CMI to the pathogen is the **Mitsuda reaction**, in which patients are challenged by intradermal injection of *M. leprae*, and induration is measured 3 weeks later. The test is positive in **tuberculoid leprosy** patients and negative in **lepromatous leprosy** patients.

It is widely accepted that **T cells** involved in CMI are pivotal in determining the outcome of *M. leprae* infection. This is supported by the observation that lymphocyte reactivity is positive in tuberculoid patients but absent in lepromatous patients. However, an interesting paradox exists: **CMI and humoral responses exhibit an inverse relationship**. Anti-*M. leprae* antibody levels are highest in patients with the **lepromatous form** of the disease and are not protective.

Cytokine Patterns in Leprosy Lesions

This paradox is best explained by the distinct **cytokine profiles** found in the two forms of leprosy:

- **Tuberculoid lesions** are characterized by strong expression of **Th1 cytokines**, particularly **IL-2** and **IFN- γ** .
- **Lepromatous lesions** are dominated by **Th2 cytokines**, including **IL-4**, **IL-5**, and **IL-10**.

These cytokine patterns correspond to the predominant T-cell subsets in the lesions:

- **CD4⁺ T cells** predominate in tuberculoid lesions. All *M. leprae*-specific CD4⁺ T cells from these patients produce **IL-2** and **IFN- γ** , and are classified as **CD4⁺ type 1 (Th1) cells**.

- **CD8⁺ T cells** predominate in lepromatous lesions. These CD8⁺ T cells produce high levels of **IL-4** and low levels of **IFN- γ** , and are designated **CD8⁺ type 2 (Tc2) cells**.

Immunopathogenesis of Leprosy

The abundance of **IL-2** and **IFN- γ** in tuberculoid lesions contributes to a robust, protective immune response:

- **IL-2** promotes clonal expansion of activated T cells and enhances **IFN- γ** production.
- **IFN- γ** activates macrophages, increasing production of reactive oxygen and nitrogen intermediates, which are critical for killing intracellular pathogens like *M. leprae*.

In contrast, the cytokine environment in lepromatous lesions promotes immune suppression:

- **IL-4** and **IL-10** drive B-cell differentiation and immunoglobulin class switching, explaining the high anti-*M. leprae* antibody levels.
- However, these cytokines also **inhibit CMI** by downregulating T-cell and macrophage functions, leading to impaired pathogen control.

Role of Innate Immunity in T-Cell Polarization

The nature of the T-cell cytokine response is influenced by the **innate immune response** (see Fig. 10-5):

- The development of a **Th1 response** is driven by **IL-12**, a 70-kD heterodimeric cytokine produced by macrophages in response to intracellular pathogens.
- IL-12 stimulates **NK cells** to produce **IFN- γ** .
- In the presence of **IL-12**, **IL-2**, and **IFN- γ** , and in the absence of **IL-4**, naïve T cells differentiate into **Th1 cells**.

- In contrast, exposure to allergens or extracellular pathogens triggers **mast cells** or **basophils** to release **IL-4**.
- In the absence of **IFN- γ** , IL-4 promotes differentiation of T cells into **Th2 cells**.

It is also intriguing to consider that **keratinocytes** may play a role in shaping the local T-cell cytokine response, though this requires further investigation.